

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

```
#include<stdio.h>

int main(){

    int n;

    scanf("%d",&n);

    int i=1,s=1;

    int counter=0;

    counter++;

    counter++;

    while(1){

        counter++;

        if(!(s<=n))

            break;

        counter++;

        i++;

        counter++;

        s+=i;

    }

    printf("%d",counter);
```

```
return 0;
```

```
}
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct